Recall, a Random Variable:

$X \sim p(x)$ ← prob. dist.

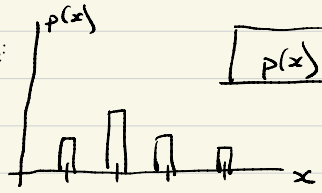
$x \in D$

E.g. $D = \{1, \dots, N\}$ discrete R.V.

or $D = \mathbb{R}$ continuous real-valued R.V.

distributions

E.g. Discrete:



$$p(x) = Pr(X = x)$$

subset $S \subset D$

E.g. $S = \{1, 3, \}$

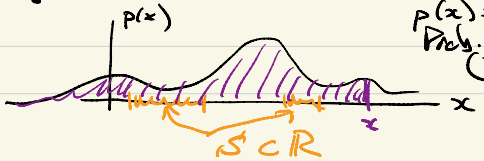
we can say:

$$Pr(X \in S) = \sum_{x \in S} p(x)$$

also:

$$\sum_{x \in D} p(x) = 1$$
$$0 \leq p(x) \leq 1$$

E.g. Continuous $D = \mathbb{R}$



$p(x)$ = differential Prob. mass of $(X = x)$.

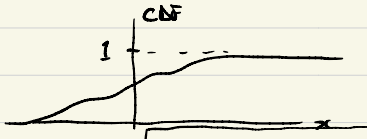
$p(x)$ is s.t.

$$Pr(X \in S) = \int_{x \in S} p(x)\,dx$$

also

$$\int_{x \in D} p(x)\,dx = 1$$
$$0 \leq p(x)$$ ← but not nec. ≤ 1

Cumulative Dist. Function (CDF):



CDF:

$$Pr(X \leq x) = \int_{-\infty}^{x} p(\xi)\,d\xi$$

notice:

$$\frac{d}{dx} Pr(X \leq x) = p(x)$$

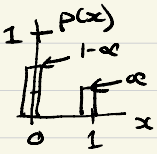
hence differential prob. mass.

Distribution Families

E.g. Bernoulli (or binary) RV

$x \in \{0, 1\}$

$$p(x) = \begin{cases} 1-\alpha & \text{for } x=0 \\ \alpha & \text{for } x=1 \end{cases}$$



one parameter: $0 \leq \alpha \leq 1$

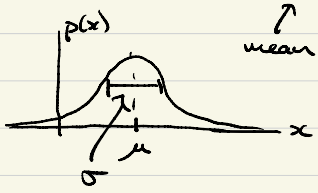
E.g. Gaussian R.V.

$x \in \mathbb{R}$

$p(x) = \mathcal{N}(\mu, \sigma^2)$ ← normal

mean, variance

σ - standard deviation



$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}}$$

two parameters: (μ, σ)

we say

$X \sim \mathcal{N}(\mu, \sigma^2)$ or $X \sim Ber(\alpha)$

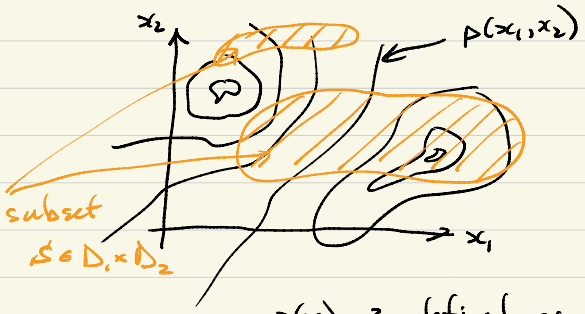
$p(x) = \mathcal{N}(\mu, \sigma^2)$ or $p(x) = Ber(\alpha)$

Jointly Distributed R.V.s

we say $(X_1, X_2) \sim p(x_1, x_2)$

and X_1 takes values $x_1 \in D_1$

X_2 takes values $x_2 \in D_2$



$p(x)$ is defined as

$$Pr((X_1, X_2) \in S) = \iint_{(x_1,x_2)\in S} p(x_1, x_2)\, dx_1 dx_2$$

(for any subset $S \in D_1 \times D_2$)
(Lebesgue measurable subset)

We can call (X_1, X_2) a 2D random vector:

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

Can extend to and n-tuple

$X = (X_1, \ldots, X_n)$ - n-dim random vector

E.g. $x \in \mathbb{R}^n$ for real-value cont. rand vector

randomness described by dist.

$p(x) = p(x_1, \ldots, x_n)$

where as before

$$Pr(X \in S) = \int_{x \in S} p(x)\, dx$$

for any (Lebesgue measurable) subset $S \subset D_1 \times \ldots \times D_n$

Jointly Dist. Random Vectors

we say $(X, Y) \sim p(x, y)$

this is equivalent to a random vector

$Z = (X, Y) \sim p(z) = p(x, y)$

choice of separating X and Y is a modeling choice

E.g. X - states
Y - measurements

Manipulating Distributions

for joint dist. R.V. (X, Y) with dist. $p(x, y)$

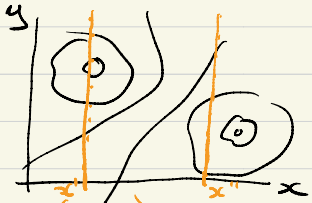
$p(x)$, $p(y)$ - marginal distributions

get from joint:

"marginalize over x (or y)" "integrate out x (or y)"

$$p(x) = \int_y p(x, y)\, dy$$

$$p(y) = \int_x p(x, y)\, dx$$



$p(x=x') \quad p(x=x'') = \int_y p(x=x'', y)\, dy$

$= \int_y p(x=x', y)\, dy$

Conditional Distributions

$(X|Y)$ or $(X|Y=y) \sim p(x|y)$

"x given y"

conditional prob. dist.

$p(x|y)$ - is a probability distribution over x (not)

$p(y|x)$ - prob. dist. over y (not x)

so

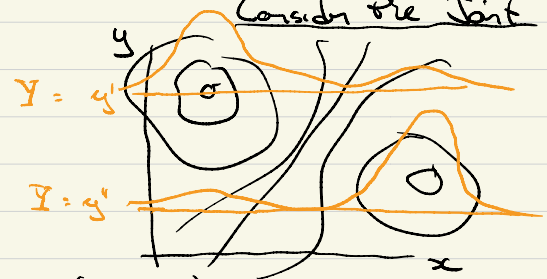
$$\int_{x \in D} p(x|y)\, dx = 1$$

$$0 \le p(x|y) \quad \forall x \in D$$

and ≤ 1 if X discrete

Intuition: Given $Y = y$, what is my new prob. mass dist. over X? what is my "belief" on X?

Consider the Joint



- $p(x, y=y')$ is correct shape on x

- but $p(x, y=y')$ does not nec. integrate to 1!
  we define conditional dist.

Def.:

$$p(x|y) = \frac{p(x,y)}{p(y)}$$

proof:

we said

$$p(x|y) = \alpha\, p(x,y)$$

find α s.t.

$$\int p(x|y)\,dx = 1$$

$$\int \alpha\, p(x,y)\,dx = 1$$

$$\alpha \int p(x,y)\,dx = \alpha p(y) = 1$$

$$\alpha = \frac{1}{p(y)}$$

so $p(x|y) = \frac{1}{p(y)} p(x,y)$ □

Factorization

$$p(x,y) = p(x|y)\,p(y)$$

also $$p(x,y) = p(y|x)\,p(x)$$

Marginalization

$$p(x) = \int_y p(x,y)\,dy$$

$$p(y) = \int_x p(x,y)\,dx$$

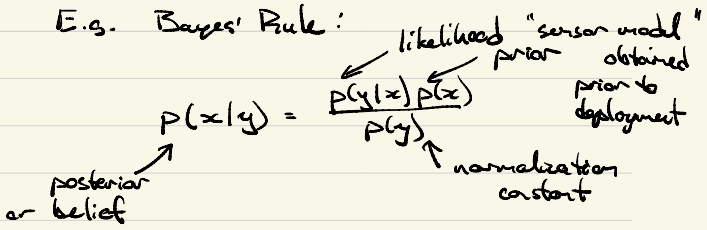
Apply these operations to manipulate distributions.

E.g. Law of total probability

$$p(x) = \int_y p(x,y)\,dy \quad \text{(marg.)}$$

$$p(x) = \int_y p(x|y)\,p(y)\,dy \quad \text{(fact.)}$$

E.g. Bayes' Rule:



proof:

$$p(x|y)\,p(y) = p(x,y) = p(y|x)\,p(x)$$

(fact.) (fact.)

$$p(x|y) = \frac{p(y|x)\,p(x)}{p(y)}$$ □

- often we know prior $p(x)$ and likelihood $p(y|x)$ but not $p(y)$
- instead we use law total prob.

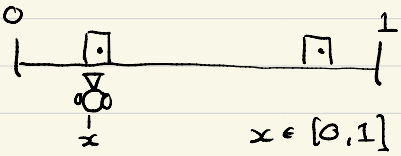
$$p(y) = \int_x p(y|x)\,p(x)\,dx$$

(likelihood, prior)

Bayes' ++

$$p(x|y) = \frac{p(y|x)\,p(x)}{\int_x p(y|x)\,p(x)\,dx}$$

E.g. robot 1-D localization



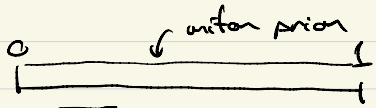
$x \in [0,1]$

$y =$  images or

 or 

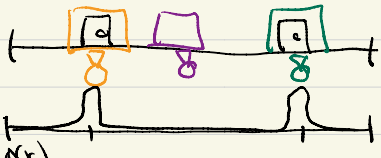
really $p(y|x)$

Bayes' Rule:

$p(x)$ 

we see $y =$ 

we know likelihood function $p(y|x)$



apply Bayes'

$p(x|y) = \frac{p(y|x)\,p(x)}{p(y)}$